

Section 16.3: The Fundamental Theorem for Line Integrals

Vector Calculus

MTH 310
Calculus III

The Big Shortcut

In Section 16.2, we computed $\int_C \vec{F} \cdot d\vec{r}$ for $\vec{F} = \langle 1 + y, x \rangle$ along the upper semicircle from $(-1, 0)$ to $(1, 0)$. It took a full page: parameterize, compute $\vec{r}'(t)$, dot product, integrate.

But $\vec{F}$ is conservative — we found in 16.1 that $f(x, y) = x + xy$ satisfies $\nabla f = \vec{F}$.

So:

$$\int_C \vec{F} \cdot d\vec{r} = f(1, 0) - f(-1, 0) = (1 + 0) - (-1 + 0) = 2$$

One line. No parameterization. And the answer is the same along **any** path from $(-1, 0)$ to $(1, 0)$.

If $\vec{F}$ has a potential function f , then $\int_C \vec{F} \cdot d\vec{r}$ reduces to plugging in endpoints. The path doesn't matter.

The Fundamental Theorem for Line Integrals

Theorem: Fundamental Theorem for Line Integrals

If $\vec{F} = \nabla f$ and C is a smooth curve from A to B , then

$$\int_C \vec{F} \cdot d\vec{r} = f(B) - f(A)$$

Compare with Calculus I:

	Calculus I	Calculus III
Integrand	$f(x) = F'(x)$	$\vec{F} = \nabla f$
Antiderivative	F	f (potential function)
Result	$F(b) - F(a)$	$f(\text{end}) - f(\text{start})$

Why? The chain rule gives $\nabla f \cdot \vec{r}'(t) = \frac{d}{dt} f(\vec{r}(t))$, so the integral collapses by the ordinary FTC.

The pattern is identical to Calc I: if you know the antiderivative, the integral depends only on the endpoints.

Example 1: Gravitational Force (Part a)

The gravitational force on a mass m due to a mass M at the origin is:

$$\vec{F}(\vec{x}) = -\frac{mMG}{|\vec{x}|^3} \vec{x} \quad \text{i.e.,} \quad \vec{F}(x, y, z) = -\frac{mMG}{(x^2 + y^2 + z^2)^{3/2}} \langle x, y, z \rangle$$

The claimed potential function is:

$$f(\vec{x}) = \frac{mMG}{|\vec{x}|} \quad \text{i.e.,} \quad f(x, y, z) = \frac{mMG}{\sqrt{x^2 + y^2 + z^2}}$$

Verify that $\nabla f = \vec{F}$.

Write $f = mMG(x^2 + y^2 + z^2)^{-1/2}$. Let $r = (x^2 + y^2 + z^2)^{1/2}$. Compute f_x using $\frac{d}{dx}r^{-1} = -r^{-2} \cdot \frac{\partial r}{\partial x}$. Then f_y, f_z by symmetry.

Example 1: Gravitational Force (Part b)

Now that we know $\vec{F} = \nabla f$ with $f(\vec{x}) = \frac{mMG}{|\vec{x}|}$:

Find the work done moving m from $(5, 10, 12)$ to $(1, 2, 3)$ along any smooth curve.

Use the FTC for line integrals: $W = f(\text{end}) - f(\text{start})$.

Three Consequences of Being Conservative

If $\vec{F} = \nabla f$ on an open, connected region D :

- 1. Path independence.** $\int_C \vec{F} \cdot d\vec{r}$ depends only on the endpoints — any path from A to B gives $f(B) - f(A)$.
- 2. Closed loops give zero.** If C starts and ends at A :

$$\oint_C \vec{F} \cdot d\vec{r} = f(A) - f(A) = 0$$

A conservative force does zero net work around any closed loop.

- 3. Cross-partials test.** If $\vec{F} = \nabla f = \langle P, Q \rangle$, then $P = f_x$ and $Q = f_y$, so Clairaut's theorem gives:

$$P_y = f_{xy} = f_{yx} = Q_x$$

On a **simply connected** domain (no holes), the converse is also true: $P_y = Q_x \Rightarrow \vec{F}$ is conservative.

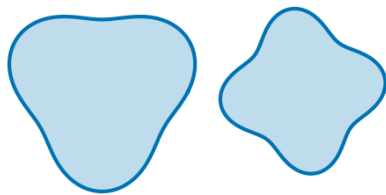
These are all equivalent: $\vec{F} = \nabla f \iff \text{path independent} \iff \oint \vec{F} \cdot d\vec{r} = 0 \iff P_y = Q_x$ (on simply connected domains).

Simply Connected Regions

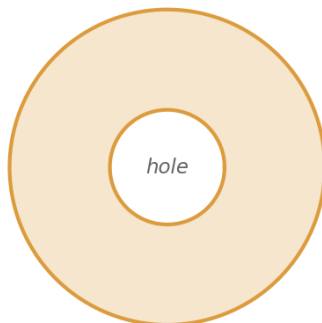
Simply Connected

A region is **simply connected** if it is one piece and has **no holes**.

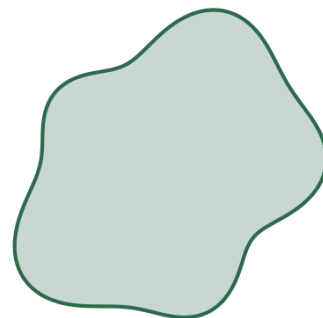
Not connected



Connected,
not simply connected



Simply connected



Why it matters: On a domain with holes, the cross-partials test can give a **false positive** — $P_y = Q_x$ everywhere, but $\vec{F}$ is still not conservative. We'll see a famous example in Section 16.4.

Quick Check: Is It Conservative?

Determine whether each field is conservative:

1. $\vec{F} = \langle 2xy + 3, x^2 - 4y \rangle$

2. $\vec{G} = \langle y^2, 3xy \rangle$

For each: identify P and Q , compute P_y and Q_x , compare.

Finding Potential Functions

When $P_y = Q_x$ confirms $\vec{F}$ is conservative, find f :

Step 1: Integrate $f_x = P$ with respect to x :

$$f(x, y) = \int P \, dx + g(y)$$

(The “constant” is a function of y — it vanishes under $\partial/\partial x$.)

Step 2: Differentiate with respect to y and set equal to Q :

$$f_y = \frac{\partial}{\partial y} \left[\int P \, dx \right] + g'(y) = Q \quad \Rightarrow \quad \text{solve for } g'(y)$$

Step 3: Integrate $g'(y)$ to find $g(y)$. Write the complete f .

You can also start by integrating $f_y = Q$ with respect to y — choose whichever integral looks easier.

Example 2: Find the Potential and Evaluate

Example 2

Show that $\vec{F} = \langle ye^x + \sin y + 3x^2, e^x + x \cos y + y^2 \rangle$ is conservative, find f , and evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, \pi/2)$.

First verify $P_y = Q_x$. Then integrate P with respect to x to find f .

Example 3: When the Potential Has No Closed Form

Example 3

Let $\vec{F} = \langle 2xy, x^2 + e^{y^2} \rangle$ and let C be the curve $y = 1 + \sin(\pi x/3)$ from $(0, 1)$ to $(3, 1)$.

Evaluate $\int_C \vec{F} \cdot d\vec{r}$.

Check: is $\vec{F}$ conservative? Try to find f . What goes wrong? Can you still evaluate the integral?

Example 4: A 3D Conservative Field

Example 4

Determine whether $\vec{F} = \langle 2xz + y^2, 2xy, x^2 + 3z^2 \rangle$ is conservative. If so, find f and evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(1, 0, 1)$ to $(2, 1, 3)$.

In 3D, if f exists then $P = f_x, Q = f_y, R = f_z$. Clairaut says all mixed partials must match: $P_y = Q_x, P_z = R_x, Q_z = R_y$. Check all three pairs.

Homework

Section 16.3: 1, 3, 5, 7, 17, 19, 21, 23, 28, 29, 31, 32, 37--40